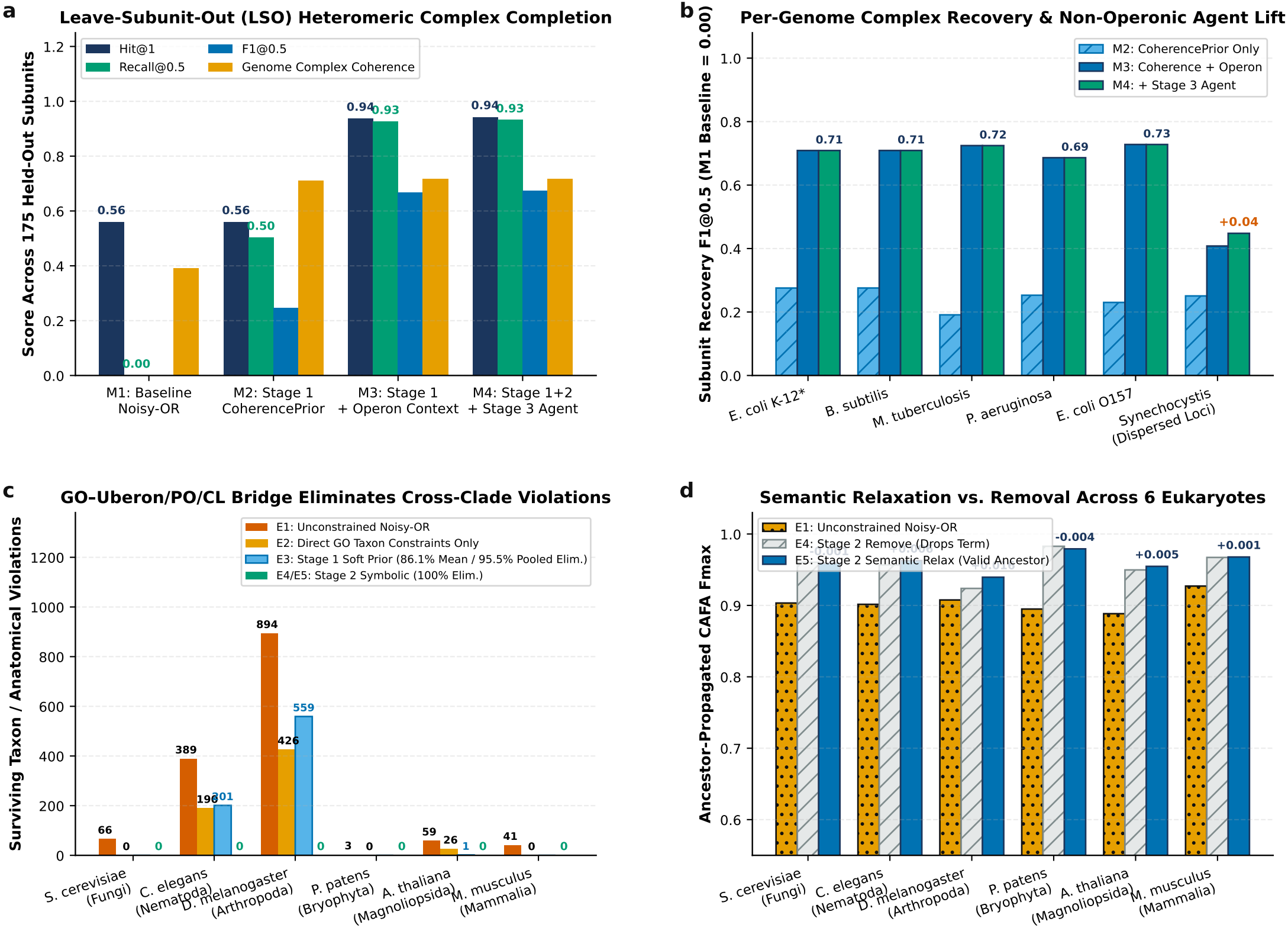

a
Leave-Subunit-Out (LSO) Heteromeric Complex Completion
Score Across 175 Held-Out Subunits
Hit@1
F1@0.5
Recall@0.5
Genome Complex Coherence
0.56
0.00
0.56
0.50
0.94
0.93
0.94
0.93
M1: Baseline Noisy-OR
M2: Stage 1 CoherencePrior
M3: Stage 1 + Operon Context
M4: Stage 1+2 + Stage 3 Agent
b
Per-Genome Complex Recovery & Non-Operonic Agent Lift
Subunit Recovery F1@0.5 (M1 Baseline = 0.00)
M2: CoherencePrior Only
M3: Coherence + Operon
M4: + Stage 3 Agent
0.71
0.71
0.72
0.69
0.73
+0.04
E. coli K-12*
B. subtilis
M. tuberculosis
P. aeruginosa
E. coli O157
Synechocystis (Dispersed Loci)
c
GO–Uberon/PO/CL Bridge Eliminates Cross-Clade Violations
Surviving Taxon / Anatomical Violations
E1: Unconstrained Noisy-OR
E2: Direct GO Taxon Constraints Only
E3: Stage 1 Soft Prior (86.1% Mean / 95.5% Pooled Elim.)
E4/E5: Stage 2 Symbolic (100% Elim.)
66
0
0
389
196
201
0
894
426
559
0
3
0
0
59
26
1
0
41
0
0
S. cerevisiae (Fungi)
C. elegans (Nematoda)
D. melanogaster (Arthropoda)
P. patens (Bryophyta)
A. thaliana (Magnoliopsida)
M. musculus (Mammalia)
d
Semantic Relaxation vs. Removal Across 6 Eukaryotes
Ancestor-Propagated CAFA Fmax
E1: Unconstrained Noisy-OR
E4: Stage 2 Remove (Drops Term)
E5: Stage 2 Semantic Relax (Valid Ancestor)
-0.004
+0.005
+0.001
S. cerevisiae (Fungi)
C. elegans (Nematoda)
D. melanogaster (Arthropoda)
P. patens (Bryophyta)
A. thaliana (Magnoliopsida)
M. musculus (Mammalia)